

Optical Fibers

Main Approaches:

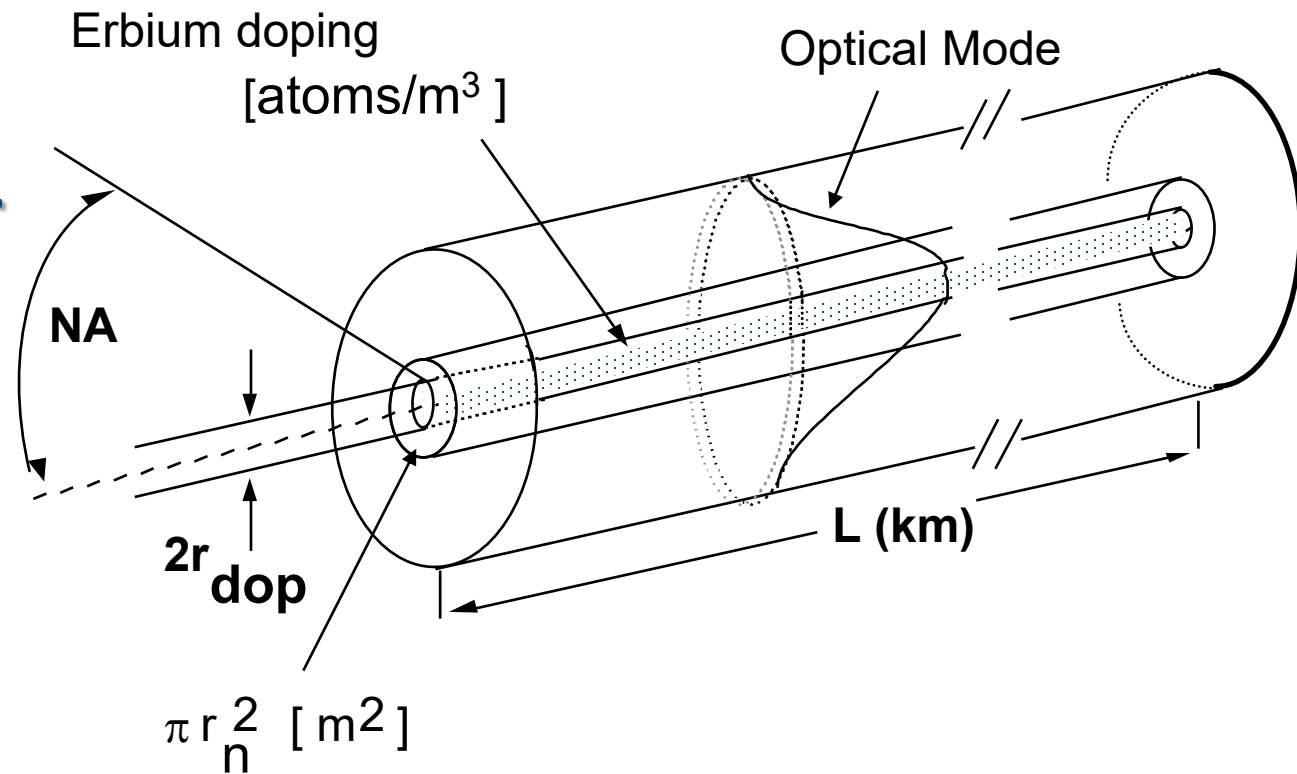
1. Geometric Optics
2. Electromagnetism: Wave Propagation

The Concept of Mode Propagation and formal description

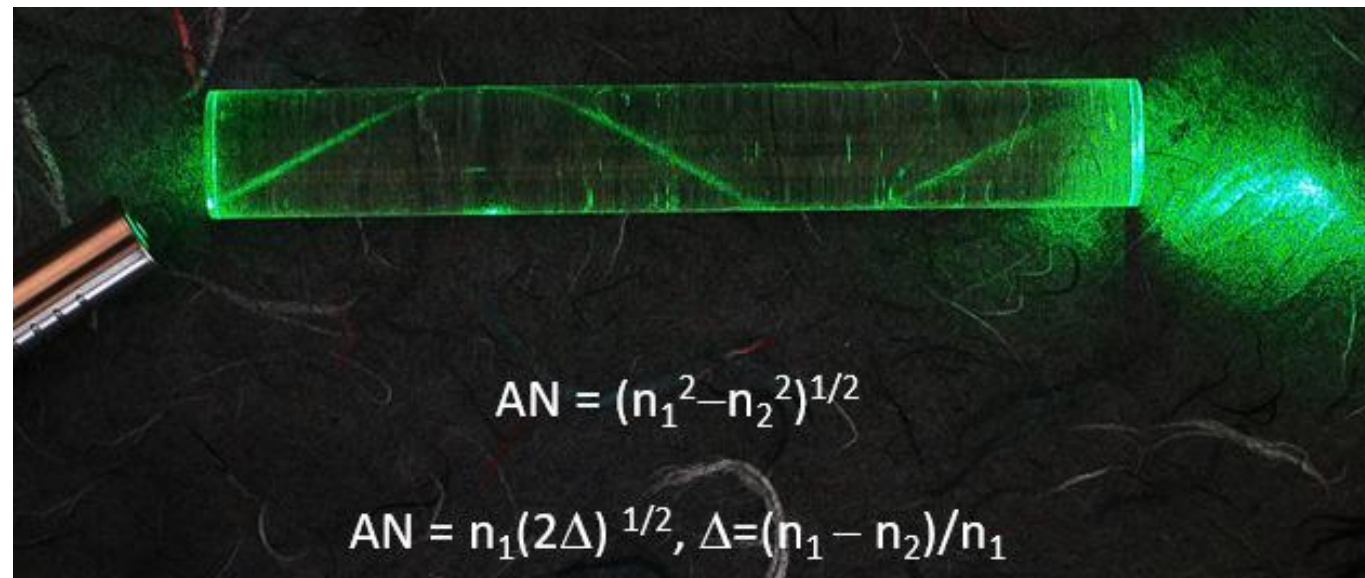
Main Equations:

- Cross-section = modes of irradiation
- Longitudinal propagation

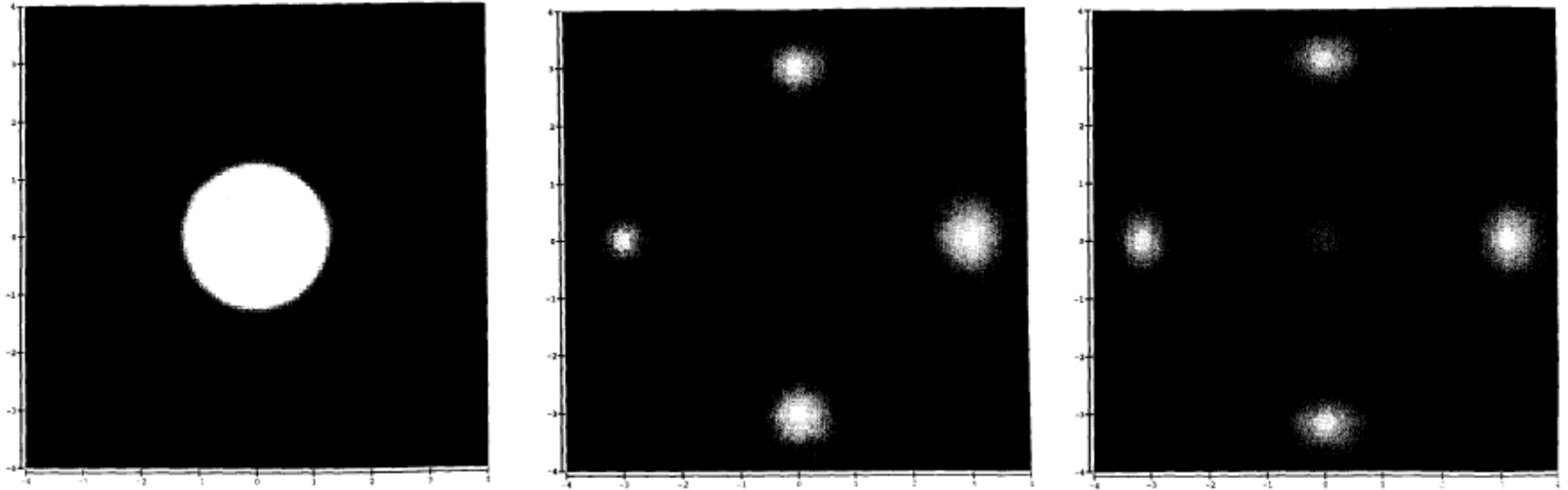
Optical Fiber



Total
internal
reflection



Visualization of Modal Irradiance Patterns in an Optical Fiber*



To determine the propagation parameters (u and w) associated with the LP modes, two nonlinear simultaneous equations must be solved. The first is the characteristic equation for the LP_{ml} modes

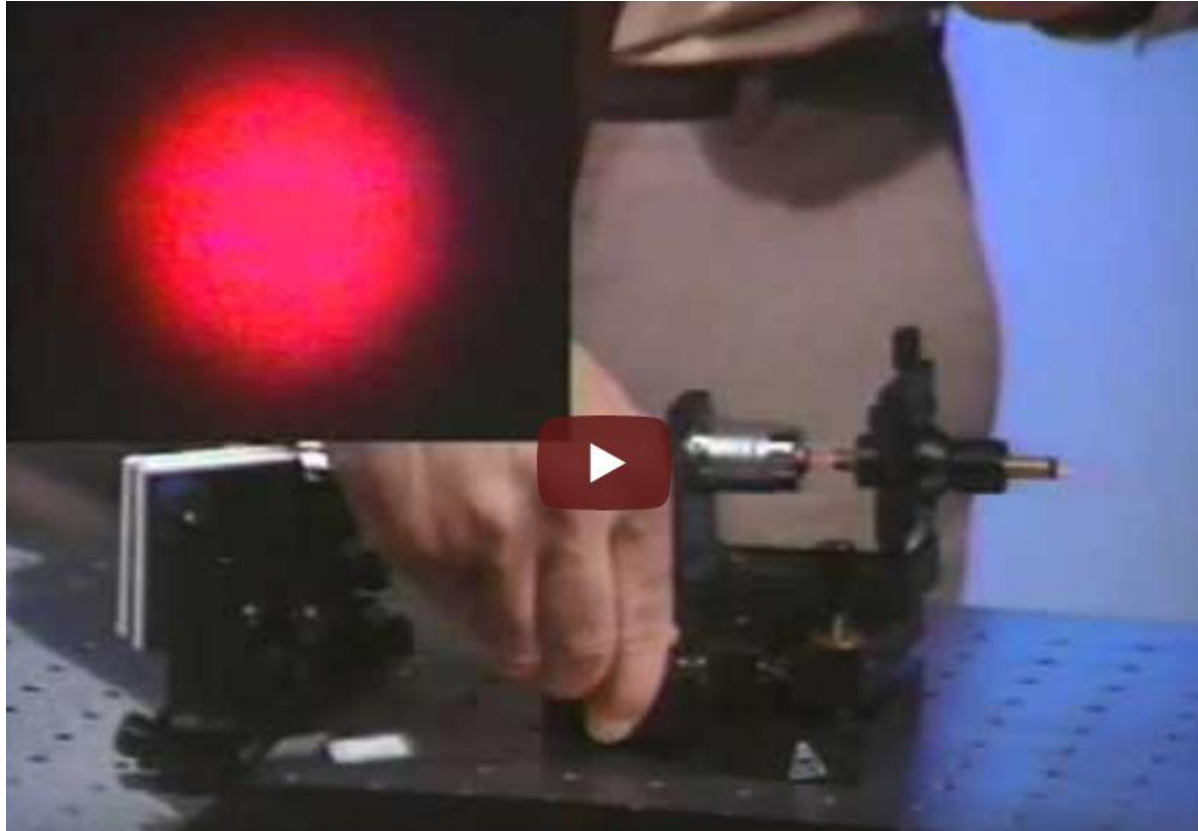
$$u^2 + w^2 = v^2. \quad (2)$$

$$(1/u)[J_m(u)/J_{m-1}(u)] = -(1/w)[K_m(w)/K_{m-1}(w)] \quad (1)$$

*Weis, "Visualization of Modal Irradiance Patterns in an Optical Fiber" IEEE Trans. on Education vol. 35, no. 2, pp. 109-111, 1992

Fiber Mode Propagation

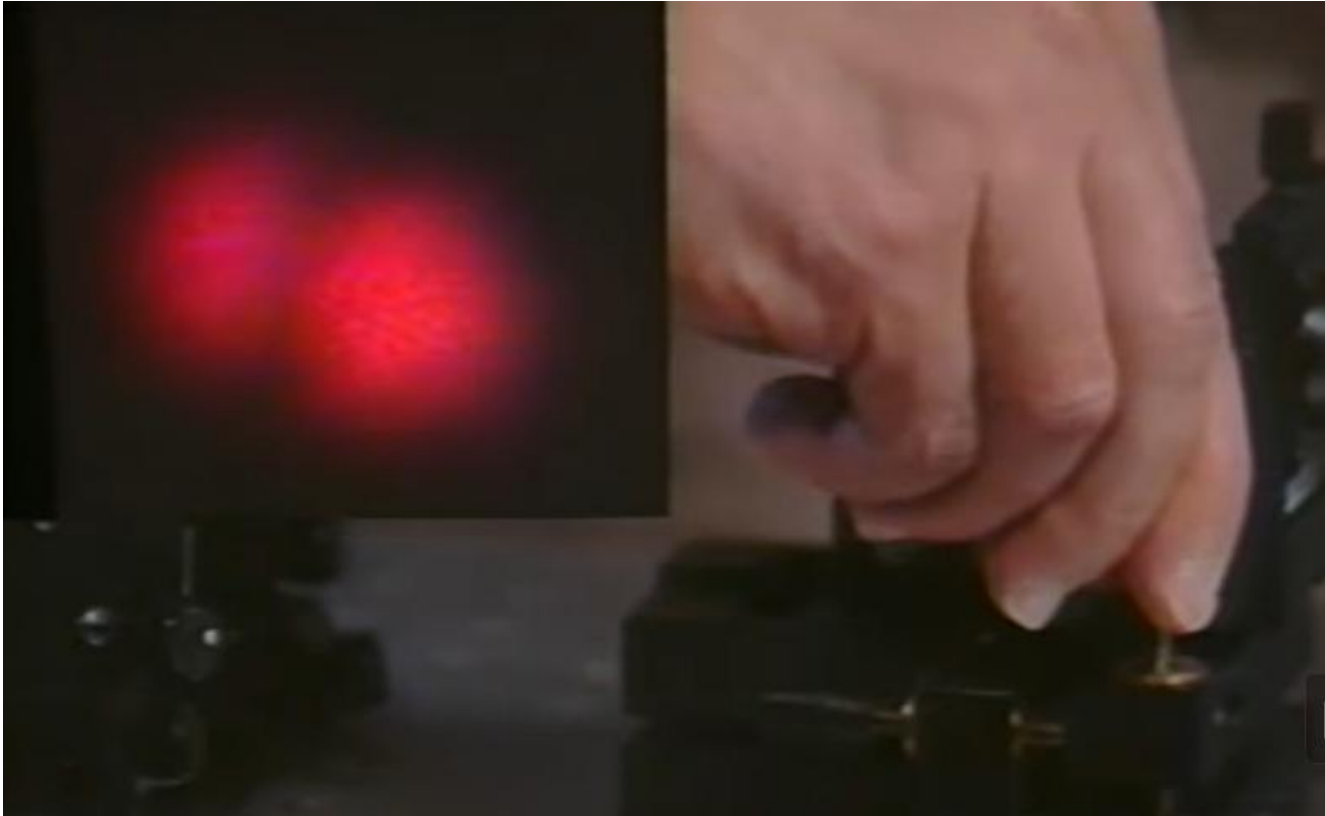
<https://www.youtube.com/watch?v=45X0puB3YK0>



Optics: Single mode fiber | MIT Video Demonstrations in Lasers and Optics

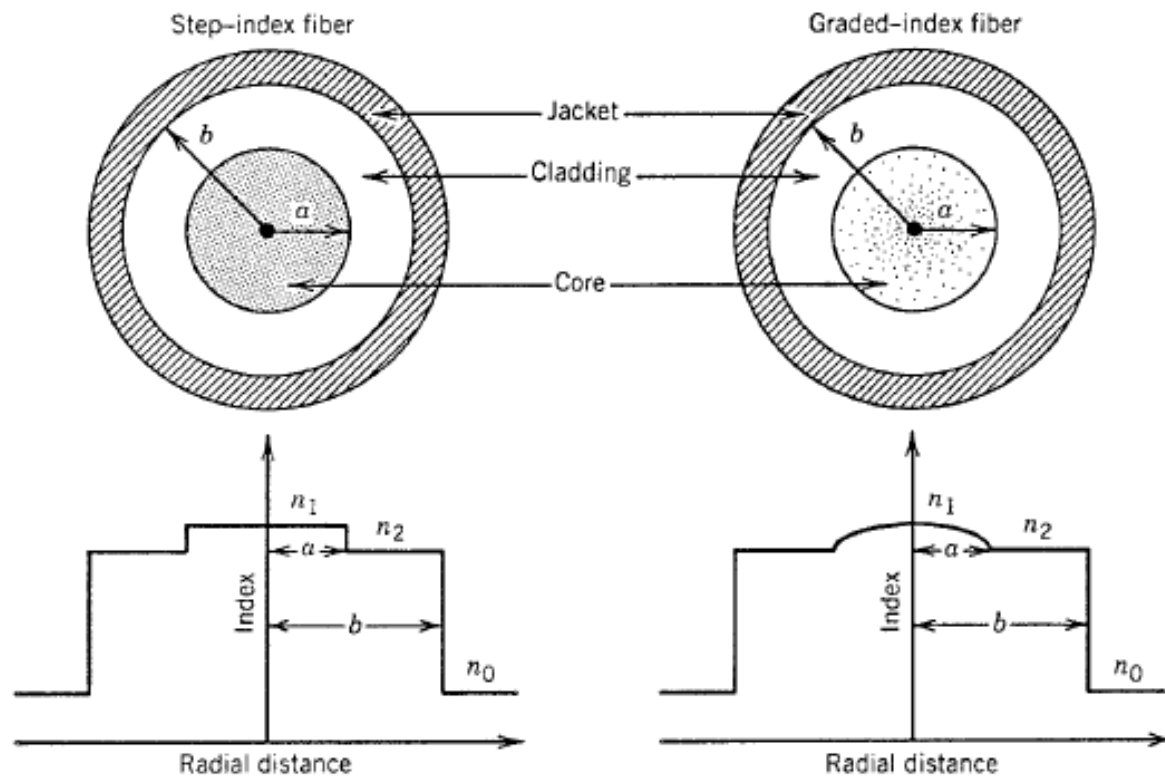
Fiber Mode Propagation

<https://www.youtube.com/watch?v=uzXLhTW9wWQ>



Optics: Multi-mode fiber | MIT Video Demonstrations in Lasers and Optics

Refractive Index Profile*



* Cap2, p. 24 - Agrawal 3^a. Ed., 2002

Figure 2.1: Cross section and refractive-index profile for step-index and graded-index fibers.

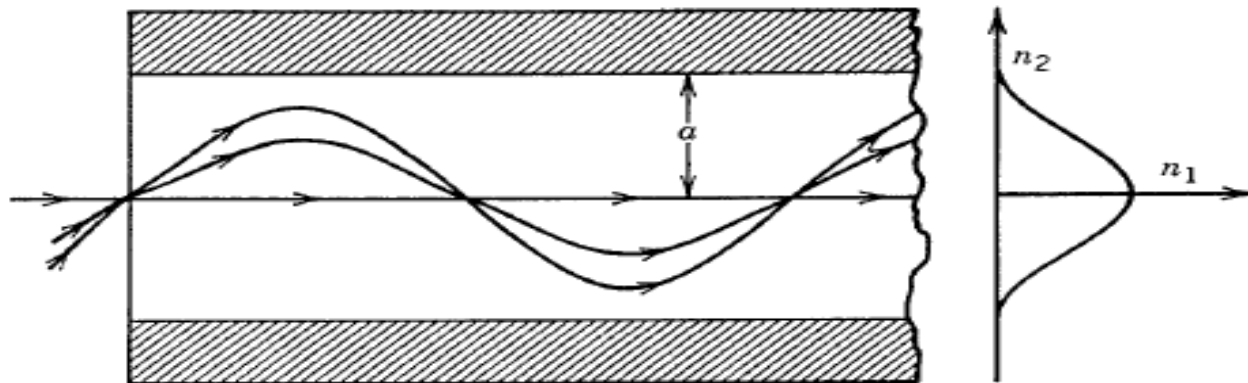
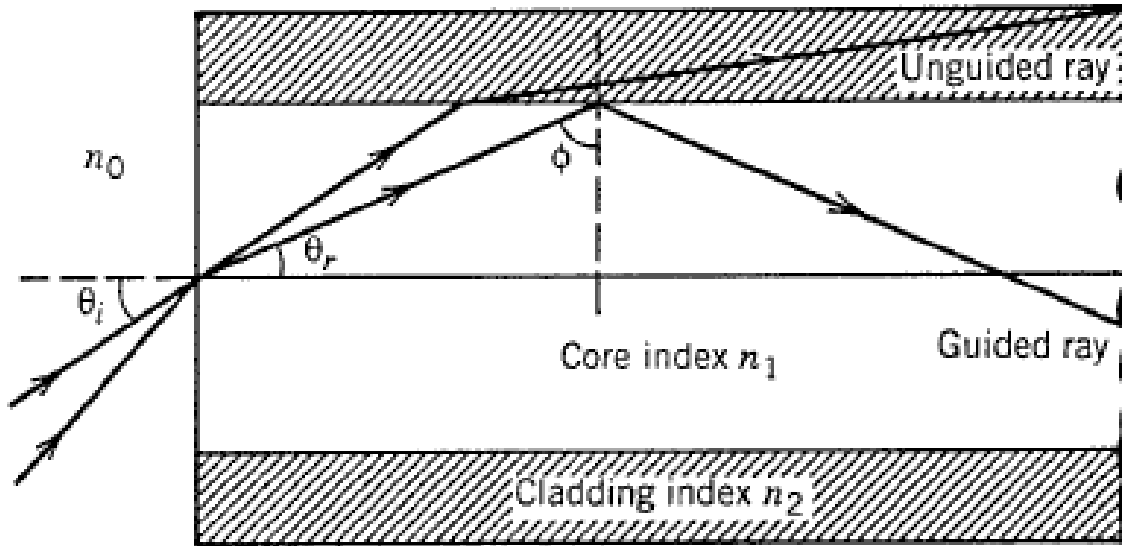


Figure 2.3: Ray trajectories in a graded-index fiber.

Refractive Index Profile*



Snell Law

$$n_0 \sin \theta_i = n_1 \sin \theta_r,$$

At the core/clad interface:

$$n_1 \sin \phi_{\text{in}} = n_2 \sin \phi_{\text{out}}$$

$$\sin \phi_c = n_2 / n_1,$$

Internal total reflexion

$$(\phi_{\text{out}} = 90^\circ)$$

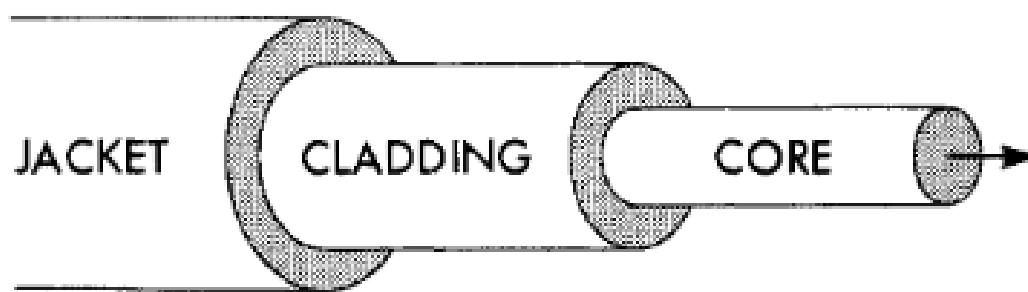
ϕ_c = critical angle

Considering,

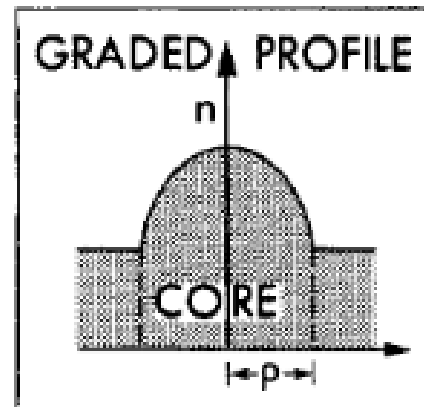
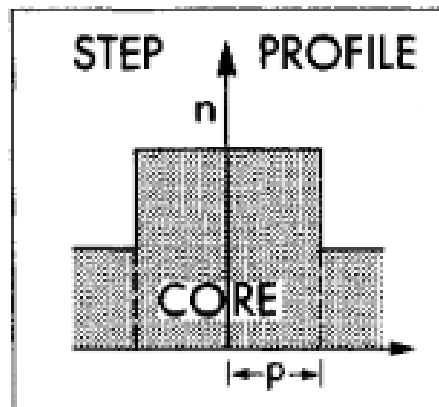
$$n_0 \sin \theta_i = n_1 \cos \phi_c = (n_1^2 - n_2^2)^{1/2}.$$

$$\theta_r = \pi/2 - \phi_c$$

$$\text{NA} = n_1 (2\Delta)^{1/2}, \quad \Delta = (n_1 - n_2) / n_1,$$



Silica Fiber



MULTIMODE FIBER

$12.5 \mu\text{m} < \rho < 100 \mu\text{m}$
 $0.8 \mu\text{m} < \lambda < 1.6 \mu\text{m}$
 $0.01 < \Delta < 0.03$

SINGLE-MODE FIBER

$2 \mu\text{m} < \rho < 5 \mu\text{m}$
 $0.8 \mu\text{m} < \lambda < 1.6 \mu\text{m}$
 $0.003 < \Delta < 0.01$

Fig. I-1 Nomenclature, profiles and ranges of dimensions for typical optical fibers, where ρ is the core radius, λ is the free-space wavelength of light and $\Delta = \{1 - n_{cl}^2/n_{co}^2\}/2$ [1].

Optical Fiber Manufacturers

Fibercore <http://fibercore.com/> (fibras especiais)

Video - <https://www.youtube.com/watch?v=ZdgEKVvlxfw>

Corning® ClearCurve ® XB Specialty Optical Fiber

<https://www.corning.com/california/products/advanced-optics/product-materials/specialty-fiber.html>

Video - https://www.youtube.com/watch?v=N_kA8EpCUQo
<https://www.youtube.com/watch?v=7tsF3mSpqX8>

Prysmian/Draka

http://prysmiangroup.com/en/business_markets/markets/telecom-solutions/products/optical-cables/

Sumitomo

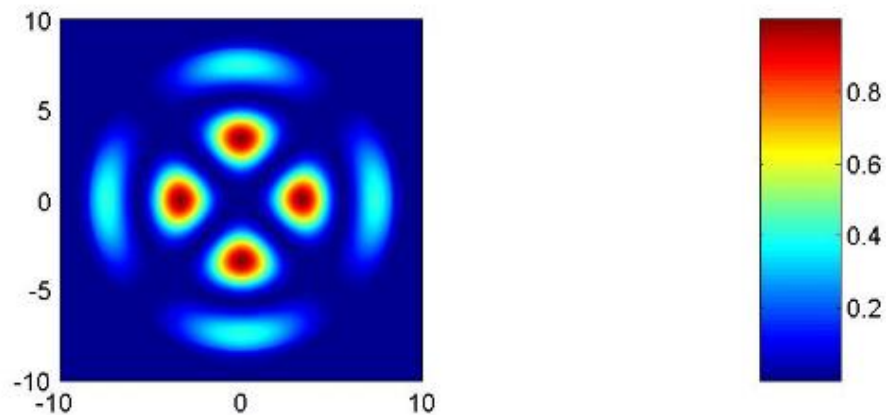
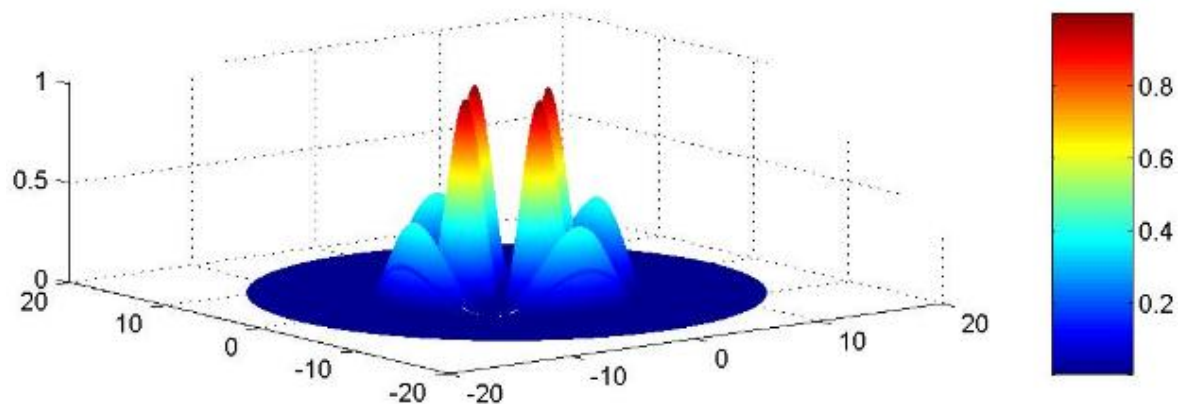
<http://www.sumitomoelectric.com/products/optical-fiber-cable/optical-fiber.html>

OFS / Furukawa

<http://www.ofsoptics.com/index.html>

Video – fiber manufacturing <https://www.youtube.com/watch?v=XvBk2l2h5Zw>

Fiber Irradiance Modes - Visualization



Exercise: Generating the irradiating modes as observed in a screen positioned at the fiber output (folder Lecturer 4: fiber modes/ matlab routines)

Wave Equations (obtained from Maxwell's Equations)

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2-27a)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \quad (2-27b)$$

$$\nabla \cdot \mathbf{D} = 0 \quad (2-27c)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2-27d)$$

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\epsilon\mu \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (2-28)$$

Using the vector identity (see App. B)

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

and using Eq. (2-27c) (that is, $\nabla \cdot \mathbf{E} = 0$), Eq. (2-28) becomes

$$\nabla^2 \mathbf{E} = \epsilon\mu \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (2-29)$$

Similarly, by taking the curl of Eq. (2-27b), it can be shown that

$$\nabla^2 \mathbf{H} = \epsilon\mu \frac{\partial^2 \mathbf{H}}{\partial t^2} \quad (2-30)$$

Onde

$$\mathbf{D} = \epsilon \mathbf{E} \text{ and } \mathbf{B} = \mu \mathbf{H}.$$

Equations (2-29) and (2-30) are the standard *wave equations*.

LP Modes & Exact Modes

